

The Impact of AI Data Centers on Electricity and Water Sustainability in Bahrain

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Course: ENGL219

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Submission Date: 14th May 2026

Table of Contents

Abstract.....	i
1. Introduction.....	1
1.1 Background Information.....	1
1.2 Purpose of the Research.....	2
1.3 Research Questions.....	2
1.4 Significance of the Research.....	3
2. Literature Review.....	4
2.1 Global Hyperscale Data Centers Energy and Water Consumption.....	4
2.1.1 The Digital Expansion and Infrastructure Growth.....	4
2.1.2 The Energy Consumption Debate.....	5
2.1.3 The Overlooked Water Footprint.....	5
2.2 The Energy-Water Relationship in the GCC.....	6
2.2.1 Water Desalination as the Backbone of Bahrain’s Municipal Water Supply.....	6
2.2.2 The Energy Cost of Water Production and Its Implications for Electricity Demand.	7
2.2.3 Environmental Consequences of Desalination and the Limits of Current Infrastructure.....	8
2.3 Technological Impacts on Resource Sustainability.....	9
2.3.1 How do AI Data Centers Drive Electricity Demand.....	9
2.3.2 Cooling Systems, Water Consumption, and the Limits of Efficiency.....	10
2.3.3 The Water Footprint of AI Systems and Its Relevance to Water-Stressed Regions..	11
3. Methodology.....	13
3.1 Research Design.....	13
3.2 Participant Selection.....	13
3.3 Data Collection.....	14
3.4 Data Analysis.....	14
4. Findings and Discussion.....	15
4.1 Electricity Consumption.....	15
4.2 Water Usage and Cooling Demands.....	16
4.3 The Energy-Water Nexus.....	17
4.4 Sustainability and Environmental Concerns.....	18
4.5 Infrastructure Readiness and Future Risks.....	19
5. Conclusion.....	20
5.1 Summary of Findings.....	20
5.2 Recommendations.....	21
6. References.....	24
7. Appendices.....	26

Abstract

When temperatures in Manama exceed 40°C, cooling the next generation of artificial intelligence becomes a national sustainability crisis rather than a technical challenge. It reveals a feedback loop in which the energy required for AI directly conflicts with the energy required to desalinate vital water supplies.

In Bahrain, the situation is worsened by a heavy reliance on energy-intensive seawater desalination and fossil-fuel energy. These factors deepen the link between energy and water, amplifying the effects of rapid data growth. As a result, this study explores how AI infrastructure affects Bahrain's national grid and water sustainability, analyzing the interaction between these resource demands and current infrastructure.

The study adopts a mixed-methods research design, combining a structured review of academic and technical literature with primary data collected through a 22-question survey administered to eight professionals at Amazon Web Services Bahrain. Quantitative responses were analyzed using descriptive statistics across the following five thematic categories: energy consumption, water usage and cooling, the energy-water nexus, sustainability and environmental impact, and infrastructure readiness. While qualitative open-ended responses were analyzed thematically.

Findings confirm that AI workloads generate electricity demands significantly higher than conventional computing operations, driven primarily by GPU-intensive processing and water-based cooling systems. Participants identified Bahrain's extreme summer climate, which regularly exceeds 40°C, as a critical amplifier of both electricity and water consumption in cooling operations. The most significant finding is the confirmation of a compounding energy-water feedback loop: because Bahrain produces virtually all of its fresh water through desalination, any increase in electricity demand driven by data center expansion directly intensifies pressure on water production capacity. This interdependence creates systematic risks that affect both sectors simultaneously. Participants also assessed Bahrain's existing infrastructure as only partially equipped to handle the concentrated loads of continued AI data center growth, and characterized the current regulatory framework as insufficiently developed to address the specific sustainability challenges these facilities pose.

The study recommends the mandatory adoption of advanced cooling technologies in new data center facilities, the development of a national data center sustainability framework requiring PUE and WUE reporting, accelerated investment in renewable energy, and the integration of energy and water resource planning into a unified national policy approach, these findings contribute to the broader academic discussion on AI infrastructure sustainability in water-stressed regions and offer actionable guidance for policymakers and operators in Bahrain and across the GCC.

1. Introduction

1.1 Background Information

Artificial intelligence has undergone rapid technological development, as its applications now extend beyond research facilities to serve as essential components of human life. People currently use three types of software: chatbot programs, graphic generation software, and coding applications. The system needs physical components, which include servers, data centers, and cooling systems, to handle each request, which results in a new answer. The technology that powers AI systems.

Data centers have always consumed vast amounts of energy. This was always an issue. AI data centers handle computing tasks that require different processing needs than traditional operations, such as website hosting and data storage. A large language model requires extensive processing capacity during training to achieve its performance targets. Model operations drive growth in energy consumption because user queries increase system demand [15].

The equation includes water as its second element, which people often overlook. Many data centers employ evaporation or water cooling for heat rejection. A typical medium-sized data center consumes many thousands of gallons of water daily for server cooling purposes. Local municipal companies provide water supply for these areas. However, this is not the case in Bahrain_[1].

Bahrain lies in the region with the highest water stress levels worldwide. The country lacks indigenous sources of fresh water. Most of the water consumed in towns and cities is obtained through desalination, a process that uses significant amounts of electricity, highlighting the interrelationship between water resources and electricity generation throughout the GCC. All the water that air conditioning

systems use needs to be replenished daily, ranging from 35 to 240 thousand liters. The operational requirements of data centers lead to their daily water use, which ranges from 35 to 240 thousand liters. Data centers operating in desert regions cannot access fresh water for their operations. Data centers incur operating expenses to secure the water supplies needed for their cooling systems. Most data centers use water-based cooling systems, which follow established standards for their operational needs. Data centers consume water through their cooling systems, which require it to maintain equipment at proper temperatures. Data centers require a continuous water supply to support their operational needs. Analytical studies show that data centers face cost challenges in maintaining their water supply operations. Our system requires municipalities to provide water because we depend on their supply for our entire cooling operations. The ongoing water supply requires an intake of between 80,000 and 300,000 liters daily to sustain our operations.

1.2 Purpose of the Research

The purpose of this study is to investigate and analyze the impact of AI data centers on electricity consumption and water sustainability in Bahrain. The study aims to examine how the energy and water demands of AI infrastructure interact with Bahrain's existing national grid and desalination-dependent water supply systems. This research combines a review of the existing academic and technical literature with primary data collected from professionals operating within AWS Bahrain to produce an evidence-based analysis of these sustainability challenges.

1.3 Research Questions

The research questions of this study are:

- RQ1: How do AI data centers in Bahrain affect national electricity consumption, and what are the primary drivers of their energy demand?
- RQ2: What is the impact of AI data center operations on water usage and cooling demands in Bahrain?
- RQ3: How does the energy-water relationship manifest in the context of AI data center expansion in Bahrain, and what risks does it create?
- RQ4: What sustainability and environmental concerns are linked to AI data center operations in Bahrain, and how ready is the existing infrastructure to accommodate continued growth?

1.4 Significance of the Research

The research conducted in this study is important because Bahrain serves as a demonstration of resource constraints that increase the difficulties faced in AI infrastructure development. The country relies almost entirely on desalination for its water supply and on fossil fuels for its electricity. The capacity of water production systems responds directly to changes in electricity usage. The research reveals local dynamics that affect both Bahrain and other Gulf Cooperation Council countries experiencing water shortages as they develop their artificial intelligence infrastructure. The study results will help decision-makers and data center managers select energy sources and cooling systems as they develop their environmental and sustainability frameworks. The research work contributes new knowledge to the academic literature by examining how the growth of AI infrastructure affects water and energy sustainability in the Gulf Cooperation Council countries. The study investigates these specific research questions. The first research question investigates how AI data centers in Bahrain impact national electricity usage.

2. Literature Review

2.1 Global Hyperscale Data Centers Energy and Water Consumption

2.1.1 *The Digital Expansion and Infrastructure Growth*

The global demand for improved connectivity solutions is driving the rapid expansion of the Information and Communication Technology (ICT) industry. Hardware requirements have reached their current level because the total number of internet-connected devices grew from 18.4 billion in 2018 to 29.3 billion by 2023, while global internet speeds reached 110 Mbps, doubling over the same period [1]. The fundamental physical infrastructure needs to undergo equivalent improvements because user numbers, internet traffic, and connection speeds have increased.

Data centers serve as the essential foundation of this infrastructure. The facilities contain millions of servers that manage, store, and relay data originating from billions of users, and the number of facilities and their storage capacity have both grown rapidly over the past 30 years [10]. The industry has moved toward large hyperscale facilities, which operate as extensive warehouse campuses that leading cloud providers such as Amazon Web Services, Google Cloud Platform, and Microsoft Azure use to run their operations. The large facilities can process much higher volumes of work while achieving better energy efficiency than the smaller facilities [1]. The increased use of AI technology leads to greater facility needs, creating challenges for researchers, policymakers, and operators who must manage these requirements.

2.1.2 The Energy Consumption Debate

The energy consumption of hyperscale data centers poses a substantial environmental problem, despite their ability to enhance digital reliability and deliver economic benefits. The data center electricity consumption worldwide for 2018 data centers reached an estimated 200 to 500 TWh range according to experts who identify ongoing challenges with both measurement techniques and industry reporting frameworks [1]. Some projections suggest that energy growth may eventually plateau as hardware efficiency improvements continue, while others warn that consumption could quadruple by 2030, driven primarily by the rising computational demands of artificial intelligence [1].

The largest operators have a clearly defined path that people increasingly agree upon. The combined electricity usage of Amazon, Microsoft, Google, and Meta more than doubled between 2017 and 2021, reaching about 72 TWh in 2021 [8]. The global data centers used approximately 460 TWh of electricity in 2022, which accounted for 2 percent of worldwide electricity demand [8]. Global data center electricity usage will increase to between 620 TWh and 1,050 TWh by 2026 because generative AI now requires businesses to invest in advanced computing power, according to the International Energy Agency, which predicts the figure will be two and a half times higher than in 2022. The ICT sector needs immediate, sustainable energy solutions and effective power management systems because its current growth rate outpaces available efficiency gains.

2.1.3 The Overlooked Water Footprint

Data center water usage needs to be studied because it is as important as energy use, yet it is rarely discussed in public and academic settings. Water is essential for industrial and agricultural activities worldwide, and forecasts indicate that total

water demand will increase by 55% between 2000 and 2050, driven by manufacturing growing by up to 400%, thermal power generation rising by 130%, and domestic consumption expanding by 140% [1]. Data centers create this demand through two methods: building cooling systems that use direct water, and generating non-renewable electricity, which requires water to operate IT equipment and cooling systems [2].

The direct water usage by major technology companies has increased dramatically in the past few years. Google's annual water withdrawal rose by approximately 28% in a single year, reaching 15.8 billion liters in 2018, while Microsoft's consumption nearly doubled over the same period, from 1.9 billion liters in 2017 to 3.6 billion liters in 2018 [1]. The total amount of water used in corporate offices includes operational expenses for data center cooling systems. The indirect water dimension adds further complexity. Water distribution systems and treatment plants consume measurable amounts of energy per million liters processed, and the embedded energy within these systems creates a feedback loop in which higher electricity demand drives higher water demand, which in turn requires additional energy to treat and distribute [2]. The relationship between these two factors has become more important now, but only about one-third of data center operators use this data.

2.2 The Energy-Water Relationship in the GCC

2.2.1 Water Desalination as the Backbone of Bahrain's Municipal Water Supply

Throughout its history, Bahrain has depended on seawater desalination as its main source of fresh water because various policy decisions, state-building efforts, and water management methods have fallen out of use. Desalination has evolved from its initial use as a temporary solution to a main water supply system that all

municipal operations now depend on for their water needs [3]. The country moved away from its existing water management methods, which included infrastructure renovation, improved tariff systems, and demand-side control, opting instead to implement a desalination-based system [3]. The development process established a system that forces Bahrain to operate with a water supply system that requires significant resources, causes severe environmental damage, and faces major challenges during necessary system changes.

The public perception of water has undergone a transformation, which serves as the main factor behind this result. The perception of water in Bahrain shifted over time from relative abundance to acute scarcity, a narrative that served to justify continued investment in desalination while deflecting scrutiny from governance failures in water demand management [3]. The existing water resource limitations and climate change pressures have transformed desalination into a political decision that governments make to support their economic plans while managing public perception of resource distribution [3]. The city relied on desalination to meet most of its water needs, creating high financial and environmental costs associated with those operations.

2.2.2 The Energy Cost of Water Production and Its Implications for Electricity Demand

The link between water supply and electricity demand in Bahrain operates through two channels that have a major impact. Research has found that approximately 8% of Bahrain's total electricity consumption is water-related, with desalination alone responsible for around 89% of that share [4]. Water production activities generate significant electricity demand, which contributes to national power needs, as Bahrain relies on fossil fuels for electricity generation [4]. The energy-water nexus

establishes both an environmental problem and a fundamental element of Bahrain's national infrastructure, which restricts large new electricity consumers from entering the nation.

Projections indicate that water demand in Bahrain is expected to rise by 27% by 2040 relative to 2019 levels, further intensifying pressure on the electricity grid [4]. Research shows that when people practice water conservation, they can achieve two major benefits: saving approximately 6,200 GWh of electricity and preventing the release of 4,300 gigagrams of CO₂ equivalent emissions [4]. These findings make a strong case for integrating water demand planning into broader energy and climate strategy in Bahrain. Future expansions by major electricity consumers, including AI data centers, require a comprehensive assessment of current and rising energy-water demands, which affect our national grid, as increased electricity demand in Bahrain drives more desalination throughout the country.

2.2.3 Environmental Consequences of Desalination and the Limits of Current Infrastructure

Desalination addresses water scarcity, but it also creates new environmental problems that harm the marine ecosystems around Bahrain. The Sitra Power and Water Station conducted research to evaluate how brine discharge from its multi-stage flash desalination plant affected nearby marine life, using hydrodynamic modeling that matched field data [5]. The research showed that brine plume temperature and salinity values returned to normal levels after 37 to 41 minutes, when the plume traveled 350 to 390 meters from the discharge area, thereby maintaining compliance with Bahrain's mixing zone regulations during the testing period [5]. The technical solution involved mixing power plant cooling water with brine discharge during the winter months, resulting in a 30 percent reduction

in temperature and a 38 percent reduction in salinity compared to current operating conditions [5].

The current operating conditions allow management of immediate thermal and salinity impacts based on study results, although the complete environmental assessment remains incomplete. The chemical composition of brine, atmospheric emissions from plant operations, and their effects on marine biological communities need to be systematically evaluated across Bahrain's desalination facilities [5]. The environmental impact on Bahrain's coastal marine environment will increase as desalination capacity expands, continuing until demand reaches its projected peak. The system's current operational capacity is constrained by environmental limits, which will be further strained by the cooling-water requirements of AI data-center operations.

2.3 Technological Impacts on Resource Sustainability

2.3.1 How do AI Data Centers Drive Electricity Demand

Artificial intelligence is expanding rapidly, transforming how data centers use electrical power. AI systems demand much greater computational resources than traditional computing systems because large language models require substantial resources for both training and ongoing inference. The transition from GPT-3 to GPT-4 at OpenAI resulted in a parameter count increase of over 900% and a training data set expansion of more than 4,000%, which caused training electricity requirements to jump from 1.29 million kWh to 50 million kWh, representing a 3,776% increase [8]. The pattern demonstrates how AI models evolve towards greater complexity, resulting in rising electricity requirements that increase with each technology cycle.

The Power distribution in a data center, which serves as its primary electrical network for operations, divides total power usage into two main categories. The Power Usage Effectiveness (PUE) system, with a value of 1.6, indicates that IT equipment consumes almost 66% of the total energy, while the cooling system uses approximately 25% [8]. The equation increases in both components as AI workloads show higher frequency and greater complexity. The International Energy Agency reports that global data center electricity use reached about 460 TWh in 2022, accounting for approximately 2 percent of worldwide energy consumption. The growing use of generative AI technologies will increase energy consumption from 620 TWh to 1,050 TWh by 2026, representing a 2.3-fold increase over the 2022 baseline [8]. The existing AI data centers will increase electricity demand on Bahrain's power system, which already faces challenges, as water desalination accounts for 8 percent of total national electricity [4].

2.3.2 Cooling Systems, Water Consumption, and the Limits of Efficiency

AI data centers require water to cool the heat generated by their server equipment. Data center facilities use cooling systems as their second-most energy-intensive equipment because most of these systems require water to operate their evaporative cooling towers and to generate the electricity they consume [2]. A 1 MW data center requires 68,000 liters of water daily for cooling [10]. The hyperscale AI facility requires this amount, which exceeds the basic operational requirements.

The cooling efficiency of a system depends on its ability to control the temperature of incoming server air. Research has shown that in the temperature range of 23 to 30°C, server power consumption increases by approximately 0.35 to 0.5% for each 1°C increase in inlet temperature because server fans must work harder to compensate for the warmer air [6]. The first option creates practical tension because

higher temperatures reduce chiller operation, which decreases cooling system costs while increasing servers' total power consumption. The ASHRAE-recommended range of approximately 25 to 27°C appears to represent a reasonable balance point for most colocation facilities, although the question of whether exceeding the upper limit yields net overall benefits remains unresolved [6]. The widely used PUE metric fails to capture this trade-off because it treats server fan consumption as IT infrastructure rather than cooling. Thus, when the temperature rises, fan energy consumption increases, total facility consumption, but PUE shows improvement. [6].

The thermal control system must address two separate challenges: controlling temperature and controlling heat generated by electronic components. Research on thermal management technologies between 2014 and 2023 shows that temperature is the main cause of over 55% of electronic device failures [7]. Heat dissipation from a system poses efficiency and reliability problems that affect the system's long-term operation. Advanced approaches, including liquid cooling, phase-change materials, and pin-fin heat pipe systems, can manage thermal loads more effectively than traditional air cooling methods, and their widespread implementation in AI data center design will lead to significant reductions in both energy and water use [7].

2.3.3 The Water Footprint of AI Systems and Its Relevance to Water-Stressed Regions

AI systems require water resources through direct and indirect use. Data centers use direct water consumption for their cooling systems, whereas their indirect water consumption comes from the water used to generate electricity for those data centers [9]. The training process of AI models requires more water than any other

current large-scale computing operation. The training process of GPT-3 required approximately 700,000 liters of water according to estimates [9]. The model uses 500 mL of water for every 20 to 50 user queries during its operational phase, resulting in rapid water consumption, as ChatGPT has surpassed 100 million users each month [9]. The total annual water consumption of AI models worldwide ranges from 3 to 6 million m³, according to broader estimates, which corresponds to the yearly water requirements of 300,000 to 600,000 people [9].

The water restrictions already affecting Bahrain make these figures important to evaluate. Bahrain relies on energy-intensive seawater desalination to meet all its municipal water needs, which are projected to increase by 27% by 2040 [4], according to section 2.2 of this document. The electricity use of AI data centers creates an indirect water footprint in Bahrain, increasing desalination requirements because the same grid-connected system is supplied by desalinated water. Research on the water footprint of data centers shows that fewer than one-third of data center operators assess their water usage, creating an industry-wide problem of insufficient transparency [1]. The absence of accountability poses a fundamental issue for all water-scarce countries seeking to develop artificial intelligence infrastructure, as every power boost requires additional water use.

The organization needs to pursue three specific strategies: implementing energy-efficient server equipment, replacing traditional cooling systems with liquid-cooling technology, designing AI model architectures that minimize unnecessary computation, and establishing water-recycling systems in its data centers [9]. The indirect water footprint decreases when organizations transition to renewable energy sources because non-renewable thermal power generation is one of the electricity production methods that requires the most water [10]. The existing

energy situation in Bahrain, which relies primarily on fossil fuels, has an indirect water impact: AI data centers require power, which directly increases desalination needs, demonstrating that water and energy requirements should be evaluated together in GCC AI infrastructure development.

3. Methodology

This section presents the research methodology employed in this study, including the research design, participant selection, data collection procedures, and data analysis techniques.

3.1 Research Design

This study adopts a mixed-methods research design, integrating both quantitative and qualitative approaches to provide a comprehensive analysis of the impact of AI data centers on electricity consumption and water sustainability in Bahrain. The quantitative component enables the measurement of participant perceptions using structured numerical data, while the qualitative component provides contextual insights to support and interpret the findings. This approach ensures both analytical rigor and interpretive depth.

3.2 Participant Selection

A total of eight participants were selected from Amazon Web Services (AWS) in Bahrain. Participants were chosen based on their direct involvement in data center operations and their familiarity with energy usage, cooling systems, and infrastructure management. To maintain confidentiality and encourage unbiased responses, all participants were anonymized and identified as Employees 1–8. Selecting participants from a single organization yields a focused case study relevant to the research context.

3.3 Data Collection

Data were collected using a structured survey instrument comprising 22 questions, developed from a previously designed 50-question framework. The survey includes 20 closed-ended questions measured using a 1–10 Likert scale and 2 open-ended questions to capture qualitative insights

The questions are organized into five thematic categories:

- (1) Energy Consumption
- (2) Water Usage and Cooling Systems
- (3) Energy–Water Nexus
- (4) Sustainability and Environmental Impact
- (5) Infrastructure Readiness and Future Risks

Each participant completed the survey individually. The Likert scale allowed for standardized quantitative evaluation, while the open-ended questions provided detailed perspectives on key challenges and future strategies.

3.4 Data Analysis

The collected data were analyzed using a combination of descriptive and qualitative thematic analyses.

Quantitative data, comprising 160 numerical responses (20 questions $\times$ 8 participants), were analyzed by calculating per-question averages and identifying trends across the five thematic categories. This enabled the identification of patterns related to energy consumption, water usage, and sustainability challenges. Qualitative responses were analyzed using thematic analysis to identify recurring concepts and key insights. These findings were used to complement and explain the quantitative results.

To ensure consistency, qualitative descriptions were systematically interpreted and aligned with the numerical scale based on predefined evaluation criteria. This integrated analytical approach enhances the validity and reliability of the study findings.

4. Findings and Discussion

This section presents the key findings from the semi-structured interviews with eight AWS Bahrain employees and discusses these findings in relation to the existing literature reviewed in Section 2. The findings are organized thematically to correspond with the research questions guiding this study.

4.1 Electricity Consumption

All eight participants unanimously confirmed that AI data centers are highly energy-intensive operations. The data center operations engineers (Employees 1–3) described the continuous and substantial electricity demands of the facility, noting that graphics processing units (GPUs) and cooling systems are the two primary drivers of energy consumption. Employee 2 explained that AI workloads, which involve complex parallel processing operations, generate significantly higher power loads than traditional computing tasks such as data storage or web hosting. The electrical engineers (Employees 4–5) further elaborated on peak load management, indicating that electricity consumption spikes during periods of intensive AI model training and real-time inference processing.

These findings are consistent with the literature, which reports that AI-optimized servers consume two to four times more energy than conventional servers ^[13] and that computing operations account for approximately 42% of total data center

power consumption [12]. Employee 5 noted that the facility employs real-time telemetry and monitoring systems to track energy usage at the server, rack, and facility levels, enabling the identification of inefficiencies and the optimization of power distribution. This practice aligns with the broader industry trend toward granular energy monitoring as a means of improving power usage effectiveness (PUE) [22]. However, despite these optimization efforts, participants acknowledged that the overall trajectory of electricity consumption is upward, driven by the increasing complexity and scale of AI applications, which is a trend corroborated by projections that global data center energy consumption will more than double by 2030 [12].

4.2 Water Usage and Cooling Demands

The mechanical engineers (Employees 6–7), who specialize in cooling systems, provided detailed insights into the water consumption patterns and cooling technologies employed at the AWS Bahrain facility. Both participants confirmed that water usage ranges from moderate to high, depending on ambient temperature conditions and workload intensity. Employee 6 explained that the facility has increasingly adopted liquid cooling technologies alongside traditional evaporative cooling systems, reflecting a broader industry shift toward more water-efficient thermal management solutions.

Bahrain's extreme climate emerged as a critical factor in cooling demands. Employee 7 reported that during summer months, when temperatures regularly exceed 45°C, cooling system energy consumption increases substantially, often requiring supplemental cooling capacity to maintain safe operating temperatures. This observation is consistent with the literature, which emphasizes that hot climates significantly amplify the cooling burden on data centers and,

consequently, their water consumption ^[15], ^[16]. The finding that 75% to 90% of data centers worldwide rely on water-based cooling ^[15] further contextualizes the challenge, as water-scarce nations such as Bahrain face inherent limitations in sustaining water-intensive cooling operations. Participants noted that water scarcity is a key operational challenge that influences facility design and technology selection decisions.

4.3 The Energy-Water Nexus

A particularly significant finding of this study is the strong confirmation of the energy-water nexus in the context of AI data centers in Bahrain. All participants recognized the intrinsic link between electricity consumption and water availability, with the site reliability engineer (Employee 8) providing an especially comprehensive perspective. Employee 8 explained that because Bahrain produces virtually all of its freshwater through desalination, any increase in electricity demand, including any that is generated by data center operations, has direct implications for the nation's water production capacity.

This finding aligns closely with the work of Al-Zubari ^[18], who reports that the water production stage consumes approximately 97% of the electricity used in Bahrain's municipal water supply cycle. The dependency creates systemic risks: if data center expansion places additional strain on the national electrical grid, the electricity available for desalination may be reduced, potentially constraining freshwater production. Several participants expressed concern about this cascading effect, noting that infrastructure strain affects both the energy and water sectors simultaneously. Qarooni ^[17] supports this assessment, highlighting that Bahrain's already limited water resources are projected to come under increasing stress, with the nation expected to rank among the top 33 water-stressed countries by 2040.

The convergence of growing AI-driven electricity demand and existing water scarcity represents a compounding sustainability risk that demands coordinated policy responses.

4.4 Sustainability and Environmental Concerns

Participants acknowledged the environmental footprint of data center operations, particularly in terms of carbon emissions. The electrical engineers noted that while the facility draws electricity from the national grid, which is predominantly powered by natural gas, AWS has implemented several measures to reduce its environmental impact. These measures include energy-efficient hardware procurement, optimization of cooling system operations, and the partial integration of renewable energy sources into the supply chain.

However, participants were candid about the difficulties of balancing sustainability objectives with operational performance requirements. Employee 3 observed that the competitive pressure to deliver low-latency, high-availability AI services often takes precedence over energy conservation measures, creating a tension between commercial imperatives and environmental goals. This observation resonates with the broader literature: while AWS, Microsoft, and Google have announced ambitious sustainability commitments [20], the pace of AI expansion continues to outstrip renewable energy deployment, resulting in a persistent gap between pledges and outcomes [19]. Guidi et al. [19] note that U.S. data centers generated over 105 million metric tons of CO₂e in 2023, with a carbon intensity that exceeds the national average by 48%. Participants also assessed the regulatory environment in Bahrain as moderately supportive, indicating that while government authorities have demonstrated awareness of sustainability issues, the policy framework has not

yet fully adapted to address the specific challenges posed by large-scale data center operations.

4.5 Infrastructure Readiness and Future Risks

The final thematic area explored in the interviews concerned Bahrain's infrastructure readiness for continued AI data center expansion. Participants provided a nuanced assessment, describing the national infrastructure as partially ready to accommodate current operations but potentially insufficient for projected growth. Employee 4, an electrical engineer, emphasized that the existing power grid was not originally designed to support the concentrated, high-density loads characteristic of data centers and that significant upgrades to generation, transmission, and distribution infrastructure would be necessary to accommodate future expansion.

The site reliability engineer (Employee 8) raised additional concerns about the cascading effects of rapid AI-driven growth on both the electrical grid and the desalination infrastructure. Employee 8 noted that without proactive investment in grid modernization and diversification of energy sources, the risk of supply constraints and their downstream impact on water production would increase materially. These observations are consistent with broader warnings in the literature about the potential for data centers to destabilize regional energy systems [21], [22]. De Vries [21] argues that the growing energy footprint of artificial intelligence necessitates urgent and coordinated action from governments, industry, and regulators to ensure that infrastructure development keeps pace with technological expansion. The participants collectively emphasized that long-term strategic planning, encompassing both energy and water resources, is essential for Bahrain to realize the economic benefits of AI adoption without compromising its environmental sustainability.

5. Conclusion

5.1 Summary of Findings

The research investigated how AI data centers affect electricity usage and water conservation efforts in Bahrain. The research produced results through a combination of an existing literature review and a structured survey that examined 8 AWS Bahrain employees across 5 thematic research areas.

The first finding establishes that AI data centers consume electricity at extremely high rates. The two primary components driving electricity consumption in AI systems are GPUs and cooling systems, as AI workloads consume more electricity than conventional computing operations [13]. The expansion of AI applications, together with their increasing complexity, results in continuous electricity usage increases despite efforts to monitor and optimize the system [12].

The second finding concerns water usage. Bahrain's summer temperatures, which exceed 45 C°, create extreme conditions that require cooling systems to operate at full capacity. Water-based cooling systems are used by 75% to 90% of data centers worldwide, according to data from [15]. This is particularly important in Bahrain, where water shortages necessitate energy-intensive desalination.

The energy-water nexus is the third finding and the most important discovery. The water supply in Bahrain depends on desalination, which produces all its freshwater needs. The water production stage alone accounts for approximately 97% of the electricity consumed in Bahrain's municipal water supply cycle [18]. The growth of AI data centers creates two problems: electricity demand and water supply

shortages. Bahrain will experience severe water shortages by 2040, which makes this problem urgent^[17].

The fourth finding relates to sustainability and environmental concerns. The pace of AI development continues to outpace the pace at which AWS and other companies have established their sustainability commitments^[20], as renewable energy resources remain insufficient to meet current demand^[19]. The participants evaluated Bahrain's regulatory framework as providing moderate assistance because it has not yet developed complete solutions for the operational needs of large data centers.

The fifth finding concerns infrastructure readiness. The original design of Bahrain's national grid did not include provisions to handle the heavy power demands of AI data centers. Future growth will require extensive upgrades to all essential systems, which include generation, transmission, and distribution networks^[21],^[22]. The electricity grid and desalination system will face increasing risks stemming from inadequate planning.

5.2 Recommendations

The study results provide policymakers, data center operators, and infrastructure planners in Bahrain with three recommended actions.

- Financial support is essential for the development of renewable energy technologies. The national grid would achieve lower carbon emissions by implementing solar and wind power, as data centers would lower their emissions, and fossil-fuel power generation would require less water.

- Organizations should use advanced cooling methods. Closed-loop and liquid-cooling systems consume water at rates below those of traditional evaporative cooling systems. The new data centers in Bahrain must implement these technologies as mandatory basic requirements.
- Bahrain needs to create a national data center sustainability framework. The country currently has no established policy framework addressing the environmental impacts of large data centers. The requirement to report power usage effectiveness (PUE) and water usage effectiveness (WUE) will enhance organizational transparency and accountability.
- The infrastructure planning process should include equipment upgrades as a preventive measure. The national power grid requires both system upgrades and capacity expansion work before AI data center growth begins to exceed its operational limits. The project needs to upgrade both transmission and distribution systems while developing multiple energy sources.
- Energy and water resource management should operate in an integrated manner. Data center expansion decisions in Bahrain need to consider their effects on desalination system capacity, as electricity use and water generation are interconnected. The country needs to adopt a national energy-water planning framework to effectively handle the increasing threats.

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7. Appendices

Appendix A – Interview Questions

The following 22 questions were used in the semi-structured interviews, organized by two thematic sections.

Section 1: Net Promoter Score (Questions 1–20)

This study uses a structured survey comprising 20 closed-ended questions to assess perceptions of AI data centers in Bahrain across five key domains: Energy Consumption, Water & Cooling, Energy-Water Nexus, Sustainability & Environment, and Readiness & Future.

A. Energy Consumption

- Q1. AI data centers are essential infrastructure for scalable intelligent computing.
- Q2. GPUs and cooling systems dominate total energy consumption.
- Q3. Electricity consumption in AI data centers is extremely high compared to most sectors.
- Q4. AI workloads significantly increase the total energy usage of a data center.
- Q5. Efficiency optimization and workload balancing are key to managing energy use.

B. Water & Cooling

- Q6. Liquid cooling is increasingly used in modern AI data centers.
- Q7. Water usage in AI data centers is moderate to high.
- Q8. Bahrain's hot climate significantly affects cooling requirements.
- Q9. Water scarcity is a key challenge for operating data centers in Bahrain.
- Q10. High temperatures increase the water demand of cooling operations.

C. Energy-Water Nexus

- Q11. Desalination plays a major role in supplying water to data centers in Bahrain.
- Q12. Energy consumption and water consumption are strongly linked.
- Q13. Desalination is critical for the continuity of data center operations.
- Q14. The interdependence between energy and water creates systematic risks.

D. Sustainability & Environment

Q15. Carbon emissions from data centers are a meaningful environmental concern.

Q16. AWS demonstrates a clear commitment to sustainability in its operations.

Q17. AI growth has a long-term environmental impact that must be addressed.

E. Readiness & Future

Q18. The rapid growth of AI data centers risks existing infrastructure.

Q19. Renewable energy is a moderately viable option in Bahrain.

Q20. Investing in sustainable infrastructure is a strategic necessity for the future.

Likert Scale Key (1 to 10)

1 — Very poor	2 — Poor	3 — Below average	4 — Slightly below average	5 — Average
6 — Slightly above average	7 — Above average	8 — Good	9 — Very good	10 — Excellent

• Net Promoter Score

Question	E1	E2	E3	E4	E5	E6	E7	E8	Average
Q1	6	4	10	8	4	6	6	8	6.50
Q2	10	4	8	4	8	6	4	6	6.25
Q3	10	6	4	6	8	4	4	10	6.50
Q4	10	6	10	4	8	4	10	8	7.50
Q5	10	8	8	8	4	4	6	10	7.25
Q6	10	4	8	6	8	10	6	4	7.00
Q7	8	6	8	8	4	6	10	4	6.75
Q8	4	4	8	8	8	6	10	8	7.00
Q9	8	8	10	4	6	6	8	6	7.00
Q10	8	8	4	10	10	8	8	4	7.50
Q11	8	4	4	10	4	10	10	6	7.00
Q12	10	4	8	6	6	8	4	4	6.25
Q13	8	6	6	8	8	8	4	4	6.50
Q14	8	8	10	4	4	10	8	4	7.00
Q15	8	8	8	10	10	10	4	6	8.00
Q16	6	10	4	8	10	8	4	8	7.25
Q17	6	8	4	8	6	10	8	10	7.50
Q18	4	8	8	8	4	4	4	8	6.00
Q19	10	6	8	6	8	8	10	8	8.00
Q20	10	8	6	8	10	8	8	6	8.00
Employee average	8.10	6.40	7.20	7.10	6.90	7.20	6.80	6.60	—

Section 2: Open-End Questions (Questions 21–22)

Employee 1

Q21 In your view, what is the biggest challenge facing AI data centers in Bahrain in terms of electricity and water sustainability?

The biggest challenge is the dual pressure on the national grid and the limited freshwater supply. AI workloads are extremely power-hungry, and every additional GPU rack translates directly into more cooling load. In Bahrain that cooling is mostly driven by desalinated water, so any growth in compute capacity quickly cascades into both higher electricity and higher water demand at the same time.

Q22 What strategic recommendations would you propose to reduce the environmental impact of AI data centers in Bahrain over the next five to ten years?

I would prioritize four moves: (1) shift heavy training workloads to off-peak hours, (2) deploy direct-to-chip liquid cooling for all new GPU clusters, (3) expand on-site solar to cover at least 25% of daytime load, and (4) sign long-term renewable PPAs with regional providers. Pairing this with stricter PUE and WUE targets would meaningfully reduce the long-term footprint.

Employee 2

Q21 In your view, what is the biggest challenge facing AI data centers in Bahrain in terms of electricity and water sustainability?

From a conservative standpoint, the challenge is real but manageable. Modern facilities are already designed with redundancy and efficiency in mind, so I would not overstate the risk. The main concern is ensuring that future expansion does not outpace the planning cycles of the electricity and water authorities.

Q22 What strategic recommendations would you propose to reduce the environmental impact of AI data centers in Bahrain over the next five to ten years?

My recommendations are incremental: continue improving PUE through better airflow management, monitor water usage with clear KPIs, and expand renewable capacity gradually rather than aggressively. Sustainability investments should follow proven technologies rather than experimental ones.

Employee 3

Q21 In your view, what is the biggest challenge facing AI data centers in Bahrain in terms of electricity and water sustainability?

The biggest challenge in Bahrain is the climate itself. Ambient temperatures regularly exceed 40 °C in summer, which forces cooling systems to run at maximum capacity for months. That directly increases both electricity draw and evaporative water loss, and there is no easy way to engineer around the local weather.

Q22 What strategic recommendations would you propose to reduce the environmental impact of AI data centers in Bahrain over the next five to ten years?

I would recommend a combination of closed-loop liquid cooling, partial use of treated wastewater for non-critical cooling stages, and stronger regional cooperation on renewable energy supply. A regional grid that shares solar and wind capacity across GCC countries would reduce the burden on any single national system.

Employee 4

Q21 In your view, what is the biggest challenge facing AI data centers in Bahrain in terms of electricity and water sustainability?

The most pressing issue is the energy-water interdependence. Producing the water we need for cooling already consumes electricity through desalination, and using that water then consumes more electricity to chill and circulate it. Any disruption on either side, whether a power outage or a water-plant issue, can cascade quickly across the data center.

Q22 What strategic recommendations would you propose to reduce the environmental impact of AI data centers in Bahrain over the next five to ten years?

In the next five to ten years I would invest in: heat recovery systems that reuse waste heat for adjacent industrial processes, hybrid cooling that combines air and liquid stages, and stricter procurement standards that require new IT equipment to meet defined efficiency benchmarks before deployment.

Employee 5

Q21 In your view, what is the biggest challenge facing AI data centers in Bahrain in terms of electricity and water sustainability?

I see the biggest challenge as the speed at which AI workloads are scaling. Capacity that used to be planned in five-year cycles is now being requested in months, and the supporting infrastructure, especially transformers, water lines, and cooling capacity, simply cannot move at the same pace.

Q22 What strategic recommendations would you propose to reduce the environmental impact of AI data centers in Bahrain over the next five to ten years?

Strategic recommendations: build modular data center sites that can be expanded in standard increments, lock in long-term renewable energy contracts now while pricing is favorable, adopt AI-driven workload scheduling to flatten demand peaks, and align early with regulators on future water-use limits.

Employee 6

Q21 In your view, what is the biggest challenge facing AI data centers in Bahrain in terms of electricity and water sustainability?

From my perspective, the toughest challenge is environmental risk perception combined with real resource limits. Bahrain has limited land, limited freshwater, and a heavily loaded grid, so any new large facility is scrutinized. AI growth amplifies all three pressures simultaneously.

Q22 What strategic recommendations would you propose to reduce the environmental impact of AI data centers in Bahrain over the next five to ten years?

I recommend doubling down on transparency and sustainability reporting, deploying renewable energy where feasible, integrating water-recycling loops into every new build, and partnering with research institutions in Bahrain to develop cooling solutions tailored to the Gulf climate.

Employee 7

Q21 In your view, what is the biggest challenge facing AI data centers in Bahrain in terms of electricity and water sustainability?

The biggest challenge is ensuring continuous operations under extreme summer conditions without overstressing the grid or the water supply. Even short interruptions in either input can force partial shutdowns, which is unacceptable for AI training workloads that may run for weeks.

Q22 What strategic recommendations would you propose to reduce the environmental impact of AI data centers in Bahrain over the next five to ten years?

I would push for: investment in large-scale solar with battery storage, use of treated greywater in cooling towers where regulations allow, deployment of high-density liquid cooling to reduce overall water footprint, and a regional resilience plan covering both electricity and water supply for critical digital infrastructure.

Employee 8

Q21 In your view, what is the biggest challenge facing AI data centers in Bahrain in terms of electricity and water sustainability?

I think the central challenge is balancing economic ambition with environmental constraints. AI is a strategic priority for Bahrain, but without careful planning, the energy and water cost of that growth could undermine the very sustainability goals the country has committed to.

Q22 What strategic recommendations would you propose to reduce the environmental impact of AI data centers in Bahrain over the next five to ten years?

My recommendations focus on governance and design: define a national sustainability framework specifically for data centers, mandate PUE and WUE reporting, require all new facilities to integrate at least one advanced cooling method such as immersion or rear-door heat exchangers, and tie incentives to measurable reductions in carbon and water intensity.

Applix B – Participants

The following table lists the anonymized participants and their professional roles.

Participant	Name	Role	Data share
Employee 1	Fatema Naser	Data Center Operations Engineer	YES
Employee 2	Anonymous	Anonymous Role	NO
Employee 3	Anonymous	Anonymous Role	NO
Employee 4	—	—	—
Employee 5	Anonymous	Anonymous Role	NO
Employee 6	Zahra Naser	Network Engineer	YES
Employee 7	Anonymous	Anonymous Role	NO
Employee 8	Mohamed Sadiq	Site Reliability Engineer (SRE)	YES

Appendix C – Interview Responses

Full interview responses from all eight participants are documented in a separate companion document titled "**AWS Bahrain Responses**". This document contains verbatim transcriptions and summarized notes from each interview session, organized by participant and thematic section. The companion document is available upon request from the research team.

Appendix D – Contributions

The following table outlines each group member's contribution to this report.

Team Member	Contribution
Alhasan Ali Abdulla	Literature review, findings and discussion, overall report compilation and editing
Ebrahim Abdulwasea Mahyoub	Introduction, methodology, data collection, and participant coordination
Hussain Ali Jaffer Shaban	Abstract, conclusion, recommendations, formatting, and references
Mohamed Habib Alhayki	Appendices, interview question design, proofreading, and quality assurance